Sreedhar Unnikrishnakurup

Senior Scientist, Structural Materials Department
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HIGHLIGHTS

- Successfully led and secured **competitive research funding** as PI and Co-PI, including a multinational grant on curved composite inspection and a collaborative aerospace composite repair project under national R&D programs.
- Published **20 journals and 37 conference papers** in leading outlets focused on thermographic inspection of composites, demonstrating impact in the field of NDT of structural materials; additionally, filed **9 technology disclosures**, with one licensed to industry partners for practical implementation.
- Over **11 years** of post-PhD experience in advanced NDT, with expertise in Infrared Thermography (Pulsed, Lock-in, Induction, and Laser-based IRT), as well as hands-on work with X-ray radiography, THz imaging, and ultrasonic inspection.
- Extensive experimental experience with sensor technologies, especially high-speed infrared cameras, and advanced NDT instrumentation for in-situ and lab-based inspection. **Level 1 certification** in Infrared Thermography.
- Proven ability to collaborate with multidisciplinary teams across academia and industry, with research experience spanning Europe and Asia.

WORK EXPERIENCE

Senior Scientist

Institute of Materials Research and Engineering (IMRE), A*STAR

Nov 2021–Ongoing — Singapore

- Project Polymer Matrix Composites Programme (PMCP): Managed Work Package that involved- In-line IRT monitoring of prepreg manufacturing, Characterization of damages CFRP [J7, C1, C9], Inspection of filament wound GFRP pipes, Impact damage characterization in CFRP using Induction IRT.
- Singapore Aerospace Program (Co-PI): UD-CFRP cross-ply laminate characterization using Induction Thermography for the composite repair [J3, J6, C3]
- Material Informed Differential Generative AI (MIDGAI) for Multi functional Composites (Project member): Development of high strength and high conductive composite using generative AI
- Other Major projects as lead researcher: NDT evaluation using IRT and Shearography of lightning strike damage in CFRP with different Lightning strike protection materials [J1], Water ingress detection in honeycomb sandwhich aircraft structures, Evaluation of coating porosity [J5], crack detection in metallic components even under paintings using induction IRT, modeling and simulation of IRT techniques and use of AI for better defect detection [J4], Infrared thermography diagnostics for reverse-bias stability in perovskite solar modules [J2]

Visiting Researcher

Mechanical Design and Machine Construction (IIND-03/A) NDT Research Group, Politecnico di Bari

November 2025 - February 2026 — Bari, Italy

- Thermal Signature Analysis for Defect Detection in Laser-Based Additive Manufacturing
- Development of Machine Learning Algorithms for the inline monitoring of LPBF process

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Senior Project Officer

Centre for Non-Destructive Evaluation (CNDE), IIT Madras

May 2018 - Sep 2021 | Chennai, India

- Application of Nondestructive IRT testing for thickness inspection of thermal barrier coatings and validation using THz-TDS and Eddy current testing [J12].
- Composite inspection using induction heating IRT. [J9]
- Crack inspection in steel billets at elevated temperatures using Laser assisted IRT [J14].
- Development of artificial intelligence-assisted advanced X-ray radiography imaging and automatic defect recognition of critical welds and high-energy materials [J10]
- Numerical investigation of anisotropic heat diffusion in polycrystalline and composite microstructures [J13].

Cohort Member BA3

Entrepreneur First

February 2020 - May 2020 | Bangalore, India

- Explored several interesting technological solutions, exploring the market, validating the technology, speaking to customers, and researching prototypes
- The ideas that we worked on is, detect and report quality-compromised and counterfeit drugs using a mobile cloud-based cost-effective hyper-spectral Imaging based AI device

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Postdoctoral Research Scientist

NDT group 8.7, Federal Institute for Material Research and Testing (BAM)

January 2016 - June 2017 | Berlin, Germany

- Project VITCEA (Validated Inspection Techniques for Composites in Energy Applications)
- Optimize and validate active thermography as full-field, fast and non-contact NDE technique for quantitative testing of FRP structures [J16]
- Laser Thermography for surface crack detection at high temperature [C20]

Visiting Research Scientist

Electromagnetic NDT Research Group, West Pomeranian University (ZUT)

January 2015 - March 2015 | Szczecin, Poland

- Scientific Exchange as part of FP7 Project: Health Monitoring of Offshore Wind Farms (HEMOW)
- Terahertz imaging and time domain analysis for material defect identification in Composite Wind turbine blades [C30]
- Active Infrared Thermography for defect identification in composite materials - Numerical investigations of heat sink effects in the infrared inspecton of composites [C26, C31]

Institute Postdoctoral Fellow

Centre for Non-Destructive Evaluation (CNDE), IIT Madras

September 2014 - December 2015 | Chennai, India

- Numerical modelling of microstructural heat diffusion and ultrasonic wave propagation in Polycrystalline Materials [C22, C24, C25]
- CMT welding process monitoring using IR thermography [J18, C18, C21]
- In-line laser thermography for Online Monitoring of steel billets to predict the surface cracks formed during the non-uniform cooling [C23, C27]

Research Associate

Centre for Non-Destructive Evaluation (CNDE), IIT Madras

April 2007 - December 2008 | Chennai, India

- Development of online weld quality monitoring for AA2219 liquid propellant tanks using Infrared Thermal Imaging and X-ray radiography [J20]
- Developed a finite element model for the prediction of temperature profiles during welding

EDUCATION

Ph.D. in Mechanics & Civil Engineering

University of Montpellier 2 (UM2)

2011 - 2014 | Montpellier, France

- Dissertation Topic: Weld pool shape identification using Multi-physics modelling and Experiments [D1, J17, C32-C36]

M.S. in Mechanical Engineering

Indian Institute of Technology Madras

2008 - 2010 | Chennai, India

- Dissertation Topic: Online weld quality monitoring using Infrared Thermal Imaging [D2, J19, J20, C29, C37]
- Cumulative Grade Point Average 8 out of 10

Bachelors in Mechanical Engineering

Institution of Engineers India

2002 - 2006 | Kottayam, India

- Cumulative Grade Point Average 6.94 out of 10

PUBLICATION

Dissertations

D1. **Sreedhar U**. *Static GTAW: experimental and numerical investigations and heat flux parameter estimation*. PhD thesis, Universite Montpellier-II, Montpellier, France, 2014. URL `https://bit.ly/2UFQyFi`

D2. **Sreedhar U**. Online weld quality monitoring using infrared thermal imaging. Master's thesis, Indian Institute of Technology Madras, Chennai, India, 2011

Journal Articles

J1. **Sreedhar U**, Santhakumar Sampath, Zi Wen Tham, Siok Wei Tay, Jonathan Zheng, Vinod Kumar, Warintorn Thitsartarn, Lei Zhang, and Andrew Ngo. Multimodal non-destructive evaluation of lightning strike damage in painted CFRP laminates: An infrared thermography-based damage index approach with cross-validation. *NDT & E International*, page 103817, 2026. link

J2. Xi Wang, Ran Luo, Nengxu Li, Jia Li, Tao Wang, Julian A. Steele, Eduardo Solano, Zihao Zhu, Xinxing Yin, Jinxi Chen, **Sreedhar U**, Andrew Ngo, Yuduan Wang, Jingcong Hu, Zhenrong Jia, Zhuojie Shi, Xinyi Du, Xiuxiu Niu, and Yi Hou. Molecular-templated pre-assembly of self-assembled monolayer for perovskite solar cells and modules with improved reverse-bias stability. *Nature Energy*, pages 1–12, 2026. link

J3. **Sreedhar U**, Renil K T Kidangan, CV Krishnamurthy, K Balasubramaniam, and Andrew Ngo. Assessment of fiber orientation and order in carbon fiber reinforced polymer composites using induction thermography and radon transform analysis. *Composites Part B: Engineering*, 296:112224, 2025. URL https://bit.ly/4c1dzsB

J4. Leith Bounenni, Mohamed Arbane, Clemente Ibarra-Castanedo, Yacine Yaddaden, **Sreedhar U**, Andrew N C Yong, and Xavier Maldague. Advanced defect detection on curved aeronautical surfaces through infrared imaging and deep learning. *NDT*, 2(4):519–531, 2024. URL https://bit.ly/4hi8sW8

J5. **Sreedhar U**, Zheng Zhang, D H Leng Seng, Zhi-qian Zhang, Jisheng Pan, Vinod Kumar, Jonathan Zheng, Carlos Manzano, and Andrew Ngo. Exploring thermal dynamics and porosity of cold-sprayed ti-6al-4v coatings on al6061-t6 substrates: A pulsed thermography and numerical modeling approach. *International Journal of Thermal Sciences*, 196:108732, 2024. URL https://bit.ly/4kDFovc

J6. Renil K T, **Sreedhar U**, CV Krishnamurthy, and Krishnan Balasubramaniam. Uncovering the hidden structure: A study on the feasibility of induction thermography for fiber orientation analysis in cfrp composites using 2d-fft. *Composites Part B: Engineering*, 269:111107, 2024. URL https://bit.ly/4igfbB6

J7. Renil K T, **Sreedhar U**, CV Krishnamurthy, and K Balasubramaniam. Induction thermography for unidirectional cfrp composites: A novel inspection approach through global current path integration. *Composite Structures*, 327:117678, 2024. URL https://bit.ly/41wBBXL

J8. Mercy L, **Sreedhar U**, J Abhinandan, M K Patra, and K Balasubramaniam. Material characterization and thickness measurement of iron particle reinforced polyurethane multi-layer coating for aircraft stealth applications using thz-time domain spectroscopy. *Journal of Infrared, Millimeter, and Terahertz Waves*, 2022. URL https://bit.ly/3vQhipj

J9. Renil Thomas Kidangan, Sreedhar Unnikrishnakurup, Chitti Venkata Krishnamurthy, and Krishnan Balasubramaniam. The influence of interlaminar microstructure on the induction heating patterns of cfrp laminates. *Materials Today Communications*, 33:104338, 2022. URL https://bit.ly/41Yc89B

J10. D Gamdha, **Sreedhar U**, and K Balasubramaniam. Automated defect recognition on x-ray radiographs of solid propellant using deep learning based on convolutional neural networks. *Journal of Nondestructive Evaluation*, 40(18), 2021. URL https://bit.ly/3aLxA7k

J11. Surendra G, Ameen E, **Sreedhar U**, K Balasubramaniam, A Veeraragavan, and B Pesala. Spectral filtering of sub-bandgap radiation using all-dielectric gratings for thermophotovoltaic applications. *Journal of Photonics for Energy*, 11(1):015501, 2021. URL https://bit.ly/3paK9hv

J12. **Sreedhar U**, J Dash, S Ray, B Pesala, and K Balasubramaniam. Nondestructive evaluation of thermal barrier coating thickness degradation using pulsed IR thermography and THz TDS measurements: A comparative study. *NDT & E*, 116: 102367, 2020. URL https://bit.ly/2GJUUHw

J13. A Saini, **Sreedhar U**, C V Krishnamurthy, T Sundararajan, and K Balasubramaniam. Numerical study using finite element method for heat conduction on heterogeneous material with varying volume fraction, shape and size of filler in matrix. *International Journal of Thermal Sciences*, 159:106545, 2020. URL https://bit.ly/31qVIYg

J14. P V Nithin, R T Kidangan, **Sreedhar U**, P Myrach, M Ziegler, and K Balasubramaniam. Laser line scanning thermography for surface breaking crack detection: modeling and experimental study. *Infrared Physics & Technology*, 104:103141, 2020. URL https://bit.ly/2YvQJ7h

J15. P V Nithin, R T Kidangan, **Sreedhar U**, P Rajagopal, K V Phani Prabhakar, G Padmanabham, and K Balasubramaniam. Numerical model and experimental validation for the on-line monitoring of cold metal transfer joining of aluminium to galvanized steel. *International Journal of Advanced Manufacturing Technology*, 104:4365–4375, 2019. URL https://bit.ly/2AnbYQF

J16. C Maierhofer, R Krankenhagen, M Roellig, **Sreedhar U**, C Monte, A Adibekyan, B Gutschwager, L Knazowicka, A Blahut, and M Gower. Influence of thermal and optical material properties on the characterization of defects in fiber reinforced composites with active thermography methods. *tm-Technisches Messen*, 85(1):13–27, 2018. URL https://bit.ly/3hswSOI

J17. **Sreedhar U**, S Rouquette, F Soulié, and G Fras. Estimation of heat flux parameters during static gas tungsten arc welding spot under argon shielding. *International Journal of Thermal Sciences*, 114:205–212, 2017. URL https://bit.ly/3d10rnl

J18. R T Kidangan, **Sreedhar U**, PV Nithin, K Balasubramaniam, P Rajagopal, KV Phani Prabhakar, G Padmanabham, F Riedel, and M Puschmann. Online monitoring of cold metal transfer (CMT) process using infrared thermography. *Quantitative InfraRed Thermography Journal*, 14(1):68–78, 2017. URL https://bit.ly/2zyzWrI

J19. **Sreedhar U**, CV Krishnamurthy, and K Balasubramaniam. Monitoring tig welding using infrared thermography–simulations and experiments. *Przegląd Elektrotechniczny*, 92(4):6–9, 2016. URL https://bit.ly/2UGtp5I

J20. **Sreedhar U**, CV Krishnamurthy, K Balasubramaniam, VD Raghupathy, and S Ravisankar. Automatic defect identification using thermal image analysis for online weld quality monitoring. *Journal of Materials Processing Technology*, 212(7): 1557–1566, 2012. URL https://bit.ly/3d5JMiA

Conference Presentations

C1. **Sreedhar U**, M. Sharma, A. Feng, J. Zheng, V. Kumar, C. Manzano, Z. Wong, and A. Ngo. Lock-in thermography and machine learning-based defect detection in carbon fiber reinforced thermoplastics. In *Singapore International NDT Conference & Exhibition*. e-Journal of Nondestructive Testing, February 2025. URL https://www.ndt.net/?id=30843. Held on 27 - 28 February 2025

C2. J Zheng, V Kumar, **Sreedhar U**, C Manzano, S Ebrahimkhani, VV Tuan, LH Anh, N-M Cheung, and ACY Ngo. Multispectral and hyperspectral infrared imaging for plastic waste sorting and recycling. In *Thermosense: Thermal Infrared Applications XLVI*, volume 13047, pages 53–62, 2024. URL https://doi.org/10.1117/12.3013731

C3. **Sreedhar U**, Renil Thomas Kidangan, CV Krishnamurthy, and Krishnan Balasubramaniam. Characterizing fiber orientations and layer stacking sequences in cfrp laminates through induction heating patterns: a nondestructive approach. In *Proceedings of 4th Asian Quantitative InfraRed Thermography Conference Characterizing*, 2023. URL https://doi.org/10.21611/qirt.2023.17

C4. S.O.A. Gnessougou, S. Unnikrishnakurup, A. Ngo Chun Yong, P. Servais, C. Ibarra-Castanedo, and X. P. V. Maldague. Curved cfrp specimen: Thermographic inspection and simulation. In *Proceedings of 4th Asian Quantitative InfraRed Thermography Conference Characterizing*, 2023. URL https://doi.org/10.21611/qirt.2023.22

C5. **Sreedhar U**, Vinod Kumar, Jonathan Zheng, Carlos Manzano, and Andrew Ngo. Exploring ambient influences on infrared thermography: A study on concrete delam-ination detection and solar loading. In *Proceedings of 4th Asian Quantitative InfraRed Thermography Conference Characterizing*, 2023. URL https://doi.org/10.21611/qirt.2023.21

C6. Vinod Kumara, Carlos Manzano, **Sreedhar U**, Jonathan Zheng, and Andrew Ngob. Dynamic inspection of surface-breaking defects using induction thermography. In *Singapore International Non-destructive Testing Conference and Exhibition (SINCE2022)*, Singapore, November 2022. URL https://doi.org/10.58286/27534

C7. Jonathan Zheng, Carlos Manzano, **Sreedhar U**, Vinod Kumar, and Andrew Ngo. Using hyperspectral imaging as a non-destructive method to discern artworks. In *Singapore International Non-destructive Testing Conference and Exhibition (SINCE2022)*, Singapore, November 2022. URL https://doi.org/10.58286/27551

C8. **Sreedhar U**, Jonathan Zheng, Vinod Kumar, Carlos Manzano, and Andrew Ngo. Crack detection in metallic components with varying surface characteristics using laser spot scanning thermography. In *Singapore International Non-destructive Testing Conference and Exhibition (SINCE2022)*, Singapore, November 2022. URL https://doi.org/10.58286/27536

C9. **Sreedhar U**, Zheng J, Manzano C, and Ngo C, Y. Numerical and experimental investigations of pulsed infrared thermography for damage detection in honeycomb sandwich composites. In *16th Quantitative Infrared Thermography Conference*, Paris, France, July 2022. URL https://doi.org/10.21611/qirt.2022.2025

C10. J Dash, **Sreedhar U**, B Pesala, and K Balasubramaniam. Material parameters and thickness determination of thermal barrier coatings using terahertz time domain spectroscopy. In *Virtual Conference and Exhibition on Nondestructive Evaluation (NDE)*, Virtual, December 2020. URL https://www.ndt.net/?id=26534

C11. Surendra G, Ameen E, A Veeraraghavan, **Sreedhar U**, K Balasubramaniam, and B Pesala. High contrast grating based broadband thermal filters for thermophotovoltaic applications. In *SPIE. Photonic West OPTO*, San Francisco, US, February 2020

C12. D Gamdha, S Dwarakanath, **Sreedhar U**, and K Balasubramaniam. Simulation assisted annotated xray image data generation for deep learning applications. In *National Seminar and Exhibition on Non-Destructive Evaluation NDE*, Bangalore, India, December 2019

C13. **Sreedhar U**, J Dash, B Pesala, and K Balasubramaniam. Non-destructive evaluation of thermal barrier coating thickness degradation using pulse phase thermography and THz-TDS. In *11th International Symposium on NDT in Aerospace*, Paris-Saclay, France, November 2019. URL https://bit.ly/2ZDVsEn

C14. S Krishna, **Sreedhar U**, and K Balasubramaniam. Evaluation of thermal barrier coating thickness using pulsed thermography experiments and numerical simulations. In *National Seminar and Exhibition on Non-Destructive Evaluation NDE*, Mumbai, India, December 2018

C15. P V Nithin, R T Kidangan, **Sreedhar U**, Krishnamurthy C, V, M Zeigler, P Myrach, and K Balasubramaniam. Numerical study of laser line thermography for crack detection at elevated temperature. In *14th Quantitative Infrared Thermography Conference*, Berlin, Germany, June 2018. URL https://bit.ly/2VLk0u8

C16. C Maierhofer, R Krankenhagen, M Röllig, **Sreedhar U**, C Monte, A Adibekyan, B Gutschwager, L Kanzowicka, A Blahut, Gower M R L, M Lodeiro, Baker G, and Aktas A. Einfluss thermischer und optischer materialeigenschaften auf die charakterisierung von fehlstellen in faserverbund-werkstoffenmit aktiven thermografieverfahren. In *Temperature*, Berlin, Germany, May 2017. URL https://bit.ly/2C7nJuS

C17. S Krishna, P V Nithin, R T Kidangan, **Sreedhar U**, M Zeigler, P Myrach, K Balasubramaniam, and Biju P. Raw data-based image processing algorithm for fast detection of surface breaking cracks. In *43rd Annual Review of Progress in Quantitative Nondestructive Evaluation, AIP Conference Proceedings*, Atlanta, Georgia, US, July 2017. URL https://bit.ly/3e2aEQW

C18. R T Kidangan, **Sreedhar U**, K Balasubramaniam, L Narayanan, and G Phanikumar. Dissimilar metal joint quality measurement using infrared thermography: Experimental and numerical approach for the application to cmt welding. In *Quantitative InfraRed Thermography*, Gdańsk, Poland, July 2016. URL https://bit.ly/3e1f8Y7

C19. P V Nithin, S Krishna, R T Kidangan, **Sreedhar U**, C V Krishnamurthy, M Zeigler, P Myrach, and K Balasubramaniam. In-line laser thermography for crack detection at elevated temperature: A numerical modeling study. In *Quantitative InfraRed Thermography*, Gdańsk, Poland, July 2016. URL https://bit.ly/2ZOCBV2

C20. B Polomski, P Myrach, E Le Claire, **Sreedhar U**, P V Nithin, K Balasubramaniam, and M Ziegler. Thermographic crack detection in hot steel surface. In *19th World Conference on Non-destructive Testing (WCNDT)*, Munich, Germany, June 2016. URL https://bit.ly/3grMvER

C21. R T Kidangan, **Sreedhar U**, P V Nithin, K Balasubramaniam, P Rajagopal, V Phani Prabhakar, K, and G Padmanabham. Application of infrared thermography technique for the monitoring of cold metal transfer (cmt) joining of aluminium to galvanized steel. In *19th World Conference on Non-destructive Testing (WCNDT)*, Munich, Germany, June 2016. URL https://bit.ly/2NYJqQp

C22. Shivaprasad S, **Sreedhar U**, S Natarajan, C V Krishnamurthy, and K Balasubramaniam. 3d elastic wave propagation and heat diffusion studies using polycrystalline material models. In *National Seminar and Exhibition on Non-Destructive Evaluation NDE*, Hyderabad, India, November 2015. URL https://bit.ly/2NYasaM

C23. **Sreedhar U**, P V Nithin, C V Krishnamurthy, M Zeigler, P Myrach, and K Balasubramaniam. The effect of surface breaking crack orientation in detection capability: A laser thermography numerical modeling approach. In *National Seminar and Exhibition on Non-Destructive Evaluation NDE*, Hyderabad, India, November 2015

C24. Shivaprasad S, **Sreedhar U**, C V Krishnamurthy, and K Balasubramaniam. Elastic wave propagation and heat diffusion studies in polycrystalline material. In *Comsol Conference*, Pune, India, October 2015. URL https://bit.ly/2BE2YqQ

C25. **Sreedhar U**, C V Krishnamurthy, and K Balasubramaniam. Heat diffusion in polycrystalline materials- a microstructure-based material model. In *Quantitative InfraRed Thermography Conference QIRT-Asia*, Mahabalipuram, India, July 2015. URL https://bit.ly/2BEf0R1

C26. B Szymznik, **Sreedhar U**, T Chady, and K Balasubramaniam. Numerical analysis of heat sink effect in the infrared inspection of composites. In *Quantitative InfraRed Thermography Conference QIRT-Asia*, Mahabalipuram, India, July 2015. URL https://bit.ly/3dZVz2m

C27. **Sreedhar U**, P V Nithinn, C V Krishnamurthy, M Zeigler, P Myrach, and K Balasubramaniam. In-line laser thermography for crack detection: A numerical approach. In *Quantitative InfraRed Thermography Conference QIRT-Asia*, Mahabalipuram, India, July 2015. URL https://bit.ly/3iwPe1F

C28. **Sreedhar U**, K Balasubramaniam, S Gollapudi, and S Kuchibhatla. Non-destructive evaluation of seams and composites at iit madras. In *Research seminar SAINT-GOBAIN University Day*, Paris, France, June 2015

C29. **Sreedhar U**, C V Krishnamurthy, and K Balasubramaniam. Monitoring tig welding using infrared thermography- simulations and experiments. In *XVIII International Symposium on Theoretical Electrical Engineering ISTET'15 & Symposium on Electromagnetic Evaluation of Materials SEEM'15*, Kolobrzeg, Poland, June 2015. URL https://bit.ly/38v5USQ

C30. T Chady, Li Yijin, and **Sreedhar U**. Evaluation of thz systems for composite structures inspection. In *XVIII International Symposium on Theoretical Electrical Engineering ISTET'15 & Symposium on Electromagnetic Evaluation of Materials SEEM'15*, Kolobrzeg, Poland, June 2015. URL https://bit.ly/3f0p9WN

C31. B Szymanik, **Sreedhar U**, and K Balasubramaniam. Background removal in thermographic non-destructive testing of composite materials. In *National Seminar and Exhibition on Non-Destructive Evaluation NDE*, Pune, India, December 2014. URL https://bit.ly/2NWtA95

C32. S Rouquette, F Soulie, and **Sreedhar U**. Gaussian heat flux parameter estimation during a gtaw operation. international conference on inverse problems in engineering. In *International conference on inverse problems in engineering*, Krakow, Poland, May 2014. URL https://bit.ly/3f32hFR

C33. **Sreedhar U**, S Rouquette, F Soulie, and G Fras. Estimation of the heat flux exchanged between argon electrical arc plasma and stainless steel anode: application to gtaw experiment. In *4th International Symposium on Inverse Problems, Design and Optimization (IPDO)*, Albi, France, June 2013. URL https://bit.ly/3dW2461

C34. **Sreedhar U**, S Rouquette, F Soulie, and G Fras. Multi-physics modeling of gtaw process and experimental validation for investigating the weld pool formation. In *International seminar Numerical Analysis of weldability*, Graz, Austria, September 2012. URL https://bit.ly/2YXEbaa

C35. **Sreedhar U**, S Rouquette, F Soulie, and G Fras. Multi-physics modeling of gtaw process. In *La Journée conference des doctorants de l'I2S (DOCTISS), The Graduate School for Information, Structures and Systems Sciences (I2S)*, Montpellier, France, June 2012

C36. S Rouquette, **Sreedhar U**, F Soulie, and G Fras. Heat flux parameter estimation by the levenberg-marquardt method: Application to the gas tungsten arc welding. In *International seminar in Inverse Problems. University of Cambridge*, London, UK, December 2011. URL https://bit.ly/2NTJbq1

C37. **Sreedhar U**, C V Krishnamurthy, K Balasubramaniam, Reghupathi V, and Ravisankar S. Modeling and simulation for temperature prediction in welding using infrared thermography. In *National Seminar NDE*, Trichy, India, December 2009. URL https://bit.ly/31KyuOw

Innovation and Technology Transfer

- Filed **9 Technology Disclosures** at Institute of Materials Research and Engineering (IMRE), A*STAR between 2021–2025 in the areas of Infrared Thermography, Composite Inspection, and Laser-Based NDT.
- **1 TD licensed** to industry partner for use in composite inspection workflows in bicycle repair.

SKILLS

●●●●● MATLAB
●●●●○ Python
●●●●● FEM: COMSOL
●●●●○ LaTeX
●●●●● Deep learning
●●●●○ Machine Learning

CERTIFICATIONS AND MEMBERSHIPS

- Project Management Fundamentals by EPM Issued on October 2022
- Infrared Thermography Level 1 by Infrared Training Center. Issued on May 2022
- Deep Learning, a 5-course specialization by deeplearning.ai on Coursera. Specialization Certificate earned on September 23, 2018
- QIRT ASIA Steering committe Member
- Member of Indian Society for Nondestructive Evaluation
- Member of Institution of Engineers (India), and Chartered Engineer Certification
- Member of NonDestructive Testing Society Singapore (NDTSS)

LANGUAGES

English ●●●●●
German ●●●○○
French ●●○○○
Malayalam ●●●●●
Hindi ●●●●○
Tamil ●●●○○

MENTORING

During postdoctoral tenure at IIT Madras: 4 Ph.D. Students and 2 Master student

- Shivaprasad S (Ph.D.) - Voronoi Cell based FEM simulations for wave propagation in solids [C22, C24]
- Renil Thomas K (Ph.D.) - Induction thermography inspection of CFRP and CMT weld monitoring [J3, J6, J9, J18]
- Nithin P V (Ph.D.) - Laser thermography for crack inspection in steel samples at elevated temperature [J14, J15]
- Surendra Gupta (Ph.D.) - Emissivity characterization of nano photonic emitters [J11]
- Anurag Saini (MTech) - Numerical investigation of Heat diffusion in particle reinforced composite [J13]
- Dhruv Gamdha (MTech) - Ray tracing model for X-ray simulation and AI based automatic defect Recognition [J10]

REFERENCES

Prof. Krishnan Balasubramaniam
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